

1. Textbook problems

6.1 Show that for a basic-state having linear shear, $\bar{U} = \lambda z$, λ constant, the basicstate QG PV depends only on z , and that the disturbance QG PV obeys

$$\left(\frac{\partial}{\partial t} + \bar{U} \frac{\partial}{\partial x} \right) q' = 0 \quad (1)$$

Ans:

The QG PV is given by

$$q = \zeta_g + f + f \frac{\partial}{\partial z} \left(\frac{d\bar{\theta}^{-1}}{dz} \bar{\theta} \right) \quad (2)$$

The basic state vorticity, $\frac{\partial \bar{U}}{\partial y}$, is zero in this situation, so the basic-state QG PV takes the form

$$\bar{q} = f + f \frac{\partial}{\partial z} \left(\frac{d\bar{\theta}^{-1}}{dz} \bar{\theta} \right)$$

Thermal wind balance for the basic state reveals that

$$\frac{\partial \bar{U}}{\partial z} = -\frac{g}{f\bar{\theta}} \frac{\partial \bar{\theta}}{\partial y} = \lambda$$

Integrating this result gives

$$\bar{\theta} = \frac{\lambda f \bar{\theta}_r y}{g} + C$$

where C is at most a function of z . Since $\zeta_g = 0$ and $\bar{\theta}$ is a function of z only, by(2),the QG PV is at most a function of z only as well.

From (6.16), Since the basic state is a function of z only, we have

$$\frac{\partial q'}{\partial t} + \mathbf{V}_g \cdot \nabla_h q' = 0$$

where primes denote perturbations from the basic state. Linearizing and noting that the basic state has only a zonal component, $\bar{U}$, gives

$$\frac{\partial q'}{\partial t} + \bar{U} \frac{\partial q'}{\partial x} = 0$$

6.2 Show that the Boussinesq continuity equation, can be written as

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\frac{\partial u_a}{\partial x_a} + \frac{\partial v_a}{\partial y} + \frac{\partial w}{\partial z} = 0$$

where the subscript a indicates ageostrophic wind.

Ans:

The total wind may be expressed as a sum of the geostrophic and ageostrophic wind component. $u = u_g + u_a$ and $v = v_g + v_a$. Plugging into the continuity equation gives

$$\frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} + \frac{\partial u_a}{\partial x} + \frac{\partial v_a}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (3)$$

Using the geostrophic wind definition.

$$u_g = \frac{1}{\rho_0 f} \frac{\partial p}{\partial y}, \quad v_g = \frac{-1}{\rho_0 f} \frac{\partial p}{\partial x}$$

gives

$$\frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} = \frac{1}{\rho_0 f} \left(\frac{\partial^2 p}{\partial x \partial y} - \frac{\partial^2 p}{\partial x \partial y} \right) = 0$$

so that (3) becomes

$$\frac{\partial u_a}{\partial x} + \frac{\partial v_a}{\partial y} + \frac{\partial w}{\partial z} = 0$$

- 6.6 (a) Derive the dispersion relationship for 3D QG Rossby waves on the β - plane. Assume plane waves in the absence of mean flow and boundaries. The governing (disturbance PV) equation is modified by an additional term

$$\frac{Dq}{Dt_g} + v\beta = 0$$

Ans:

The linearized governing equation is given by

$$\frac{\partial q'}{\partial t} + v'\beta = 0 \quad (4)$$

Assuming the wave solution form for the perturbation pressure,

$$p' = \text{Re}\{Ae^{i(kx+ly+mz-vt)}\} \quad (5)$$

and using (6.24) gives

$$\frac{\partial q'}{\partial t} = \frac{i\nu}{\rho_0} \left(k^2 + l^2 + \frac{f^2}{N^2} m^2 \right) p' \quad (6)$$

From the geostrophic wind relationship

$$v' = \frac{ik}{\rho_0 f} p'$$

Substituting (5) and (6) in (4) gives

$$\frac{i\nu}{\rho_0 f} \left(k^2 + l^2 + \frac{f^2}{N^2} m^2 \right) p' + \frac{ik\beta}{\rho_0 f} p' = 0$$

Solving for ν gives

$$\nu = \frac{-\beta k}{k^2 + l^2 + \frac{f^2}{N^2} m^2}$$

- (b) Derive the group velocity vector. Explain the result.

Ans:

Group velocity is defined by $\mathbf{C}_g = \nabla_K \nu = \left(\frac{\partial \nu}{\partial k} + \frac{\partial \nu}{\partial l} + \frac{\partial \nu}{\partial m} \right)$. Let

$$K = k^2 + l^2 + \frac{f^2}{N^2} m^2$$

The scalar components of the group velocity are then given by

$$\frac{\partial \nu}{\partial k} = -\frac{\beta k}{kK} + 2\frac{k^2 \beta}{K^2} = \frac{\nu}{k} - 2\nu \frac{k}{K}$$

$$\frac{\partial \nu}{\partial l} = \frac{\beta k}{K^2} 2l = -\nu \frac{l}{K}$$

$$\frac{\partial \nu}{\partial m} = \frac{\beta k}{K^2} 2m = -\nu \frac{m}{K}$$

Therefore,

$$\mathbf{C}_g = -2\frac{\nu}{K} \left(k - \frac{K}{2k}, l, m \right)$$

The group velocity is directed opposite to the phase speed, except when $k - \frac{K}{2k} < 0$, or

$$k^2 < l^2 + \frac{f^2}{N^2} m^2$$

which applies to waves with sufficiently long wavelengths in the zonal direction compared to the meridional and vertical directions.

- 6.7 Derive the QG kinetic energy equation

$$\frac{\partial K}{\partial t_g} + \nabla \cdot \vec{S} = \frac{g}{\theta_0} w\theta$$

starting with the QG momentum equation

$$\frac{D\mathbf{V}_g}{Dt_g} = -f\mathbf{k} \times \mathbf{V}_a$$

Ans:

The QG kinetic energy is defined as usual by

$$K = \frac{1}{2} \mathbf{V}_g \cdot \mathbf{V}_g = \frac{1}{2} (u_g^2 + v_g^2)$$

Taking $\mathbf{V}_g \cdot$ (momentum equation), gives

$$\mathbf{V}_g \cdot \frac{D\mathbf{V}_g}{Dt_g} = \frac{DK}{Dt_g} = -f \mathbf{V}_g \cdot (\mathbf{k} \times \mathbf{V}_a)$$

Expanding the right side gives

$$\frac{DK}{Dt_g} = f v_a u_g - f u_a v_g = -\frac{1}{\rho_0} \frac{\partial}{\partial y} (v_a p) - \frac{1}{\rho_0} \frac{\partial}{\partial x} (u_a p) + \frac{p}{p_0} \left(\frac{\partial u_a}{\partial x} + \frac{\partial v_a}{\partial y} \right)$$

where the geostrophic wind relationships were used in the last equality. Using the QG continuity equation gives

$$\frac{DK}{Dt_g} = -\frac{1}{\rho_0} \nabla_h \cdot (u_a p + v_a p) - \frac{p}{p_0} \frac{\partial w}{\partial z}$$

Using the hydrostatic equation on the last term yields

$$\frac{DK}{Dt_g} = -\nabla \cdot \mathbf{F}_a + \frac{g}{\theta} w \theta \quad (7)$$

where $F_a \equiv \frac{1}{\rho_0} (u_a p, v_a p, w p)$. Due to the nondivergence of the geostrophic wind, $\mathbf{V}_g \cdot \nabla_h K = \nabla_h \cdot (\mathbf{V}_g K) \equiv \nabla_h \cdot \mathbf{F}_g$. Letting $\mathbf{S} = \mathbf{F}_a + \mathbf{F}_g$ allows (7) to be written as

$$\frac{\partial K}{\partial t} = -\nabla \cdot \mathbf{S} + \frac{g}{\theta} w \theta$$

which shows that the local change in QG kinetic energy is due to a conversion of potential energy ($w\theta$), positive for potentially cold air sinking and potentially warm air rising, and a flux divergence associated with flux vector $\mathbf{S}$. Part of the flux relates to the geostrophic advection of K , and another part to the ageostrophic flux of pressure.

- 8.1 Verify by direct substitution that the Ekman spiral expression (8.26) is indeed a solution of the boundary layer equations (8.21) and (8.22) for the case 0. $v_g = 0$

$$K_m \frac{\partial^2 u}{\partial z^2} + f(v - v_g) = 0 \quad (8.21)$$

$$K_m \frac{\partial^2 v}{\partial z^2} + f(u - u_g) = 0 \quad (8.22)$$

$$u = u_g(1 - e^{-\gamma z} \cos \gamma z), \quad v = u_g e^{-\gamma z} \sin \gamma z \quad (8.26)$$

Ans:

From (8.26), we have

$$\frac{\partial^2 u}{\partial z^2} = -2u_g \gamma^2 e^{-\gamma z} \sin \gamma z, \quad \frac{\partial^2 v}{\partial z^2} = -2u_g \gamma^2 e^{-\gamma z} \cos \gamma z$$

, and substituting into (8.26) and (8.27) gives $-2\gamma^2 K u_g + f u_g = 0$. Hence, $\gamma^2 = f/2K$, so solution checks.

- 8.2 Derive the Ekman spiral solution for the more general case where the geostrophic wind has both x and y components (u_g and v_g , respectively), which are independent of height.

Ans:

In this case the general solution of (8.20) is

$$u + iv = A \exp[(if/K)^{1/2} z] + B \exp[-(if/K)^{1/2} z] + (u_g + iv_g)$$

Applying the boundary conditions (8.28) for $f > 0$ gives $A = 0$ and $B = -(u_g + iv_g)$. Taking the real and imaginary parts yields the velocity components in the Ekman layer:

$$u = u_g(1 - e^{-\gamma z} \cos \gamma z) - v_g e^{-\gamma z} \cos \gamma z$$

$$v = v_g(1 - e^{-\gamma z} \cos \gamma z) + u_g e^{-\gamma z} \cos \gamma z$$

- 8.5 Suppose that at 43°N the geostrophic wind is westerly at 15 m s⁻¹. Compute the net cross-isobaric transport in the planetary boundary layer using both the mixed layer solution (8.17) and the Ekman layer solution (8.26). You may let $|\mathbf{V}| = u_g$ in equation (8.17), $h = D_e = 1$ km, $\kappa_s = 0.015$ m⁻¹ s, and $\rho = 1$ kg m⁻³.

$$\bar{v} = \kappa_s |\mathbf{V}| \bar{u}; \quad \bar{u} = \bar{u} - \kappa_s |\mathbf{V}| \bar{s} \quad (8.17)$$

$$u = u_g(1 - e^{-\gamma z} \cos \gamma z), \quad v = u_g e^{-\gamma z} \sin \gamma z \quad (8.26)$$

Ans:

For the mixed layer case the cross isobaric mass flux is $M = \rho \bar{v} h \text{ kg m}^{-1}$. But from (8.22)

$$\bar{v} = \kappa_s |\bar{\mathbf{V}}| \frac{u_g}{1 + \kappa_s^2 |\bar{\mathbf{V}}|^2} = \frac{\kappa_s u_g^2}{1 + \kappa_s^2 u_g^2} = 7.2 \text{ ms}^{-1}$$

,so mixed layer case gives $M = \rho \bar{v} h = 7.2 \times 10^3 \text{ kg m}^{-1} \text{ s}^{-1}$. For the Ekman layer, $\gamma = \pi/D_e$, so

$$M = \int_0^{D_e} \rho v dz = \int_0^{D_e} \rho u_g e^{-\gamma z} \sin \gamma z dz = \rho u_g (2\gamma)^{-1} (1 + e^{-\pi}) = 2.49 \times 10^3 \text{ kg m}^{-1} \text{ s}^{-1}$$

- 8.6 Derive an expression for the wind-driven surface Ekman layer in the ocean. Assume that the wind stress τ_w is constant and directed along the x axis. The continuity of turbulent momentum flux at the air-sea interface ($z = 0$) requires that the wind stress divided by air density must equal the oceanic turbulent momentum flux at $z = 0$. Thus, if the flux-gradient theory is used, the boundary condition at the surface becomes

$$\rho_0 K \frac{\partial u}{\partial z} = \tau_w, \quad \rho_0 K \frac{\partial v}{\partial z} = 0, \quad \text{at } z = 0$$

where K is the eddy viscosity in the ocean (assumed constant). For a lower boundary condition, assume that $u, v \rightarrow 0$ as $z \rightarrow -\infty$. If $K = 10^{-3} \text{ m}^2 \text{ s}^{-1}$, what is the depth of the surface Ekman layer at 43°N latitude?

Ans:

For $u_g = v_g = 0$, the general solution of (8.20) is

$$u + iv = A \exp[(if/K)^{1/2} z] + B \exp[-(if/K)^{1/2} z]$$

But, $u + iv \rightarrow 0$ as $z \rightarrow -\infty$, which gives $B = 0$. From the surface boundary condition,

$$\left(\frac{\partial u}{\partial z} \right)_{z=0} = \frac{\tau_w}{\rho K} = \text{Re}[A(if/K)^{1/2}], \quad \left(\frac{\partial v}{\partial z} \right)_{z=0} = 0 = \text{Im}[A(if/K)^{1/2}]$$

But, $\sqrt{i} = [\exp(i\pi/2)]^{1/2} = \exp(i\pi/4) = (1+i)/\sqrt{2}$.

Thus, the condition on $(\partial v/\partial z)z = 0$ implies that $A_{re} = -A_{im}$, while the condition on $(\partial u/\partial z)z = 0$ implies that $A_{re} = (\tau_w/2\rho K\gamma)$, where $\gamma = (f/2K)^{1/2}$.

$$\therefore u = A_{re} e^{\gamma z} (\cos \gamma z + \sin \gamma z); \quad v = A_{re} e^{\gamma z} (-\cos \gamma z + \sin \gamma z)$$

Here $D_e = (\pi/\gamma) = \pi(2 \times 10^{-3}/10^{-4})^{1/2} = 14.05 \text{ m}$

- 8.7 Show that the vertically integrated mass transport in the wind-driven oceanic surface Ekman layer is directed 90° to the right of the surface wind stress in the Northern Hemisphere. Explain this result physically.

Ans:

The only forces acting on the water column are the wind stress and the Coriolis force. Thus, the Coriolis force on the water column must oppose the wind stress vector, so the mass transport must be 90° to the right of the surface wind stress. This can be verified by considering the solution to Problem 8.6:

$$\int_{-\infty}^0 \rho u dz = \int_{-\infty}^0 \rho A_{re} e^{\gamma z} (\cos \gamma z + \sin \gamma z) dz = 0$$

$$\int_{-\infty}^0 \rho v dz = \int_{-\infty}^0 \rho A_{re} e^{\gamma z} (-\cos \gamma z + \sin \gamma z) dz = -(\tau_w/f) \text{ kmm}^{-1} \text{ s}^{-1}$$

- 8.10 Derive a formula for the vertical velocity at the top of the planetary boundary layer using the mixed layer expression (8.17). Assume that $|\bar{\mathbf{V}}| = 5 \text{ ms}^{-1}$ is independent of x and y and that . If $h = 1$ km and $\kappa_s = 0.05$, what value must K_m have if this result is to agree with the vertical velocity derived from the Ekman layer solution at 43° latitude with $D_e = 1 \text{ km}$?

$$\bar{v} = \kappa_s |\mathbf{V}| \bar{u}; \quad \bar{u} = \bar{u} - \kappa_s |\mathbf{V}| \bar{s} \quad (8.17)$$

Ans:

From(8.22)

$$\bar{u} = \bar{u}_g(1 + \kappa_s^2 |\mathbf{V}|^2)^{-1}; \quad \bar{v} = \kappa_s |\mathbf{V}| \bar{u}_g(1 + \kappa_s^2 |\mathbf{V}|^2)^{-1}$$

,and from the continuity equation

$$w(h) = - \int_0^h \left(\frac{\partial \bar{v}}{\partial y} \right) dz = \frac{-\kappa_s |\mathbf{V}| h}{(1 + \kappa_s^2 |\mathbf{V}|^2)} \frac{\partial \bar{u}_g}{\partial y} = \frac{-\kappa_s |\mathbf{V}| h}{(1 + \kappa_s^2 |\mathbf{V}|^2)} \bar{\zeta}_g$$

For equality with the Ekman layer result, (8.38) must have

$$\frac{-\kappa_s |\mathbf{V}| h}{(1 + \kappa_s^2 |\mathbf{V}|^2)} = \left(\frac{K_m}{2f} \right)^{1/2}$$

Solving for K_m gives $K_m = 11 \text{ m}^2 \text{s}^{-1}$.

2. Inclass problems.

1. With the tendency equation shown below, please derive and interpret the physical meaning of the diabatic term.

$$\left(\nabla_h^2 + \frac{f_0^2}{\sigma} \frac{\partial^2}{\partial p^2} \right) \chi = -f_0 \vec{V}_g \cdot \nabla (\zeta_g + f) + \frac{f_0^2}{\sigma} \frac{\partial}{\partial p} \left[\vec{V}_g \cdot \nabla \left(-\frac{\partial \Phi}{\partial p} \right) \right] - \frac{f_0^2}{\sigma} \frac{\partial Q}{\partial p}$$

Ans:

From derivative of vorticity and β -plane approximation

$$\begin{aligned} \frac{\partial \zeta_g}{\partial t} &= -\vec{V}_g \cdot \nabla (\zeta_g + f) + (\zeta_g + f_0) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \\ &= -\vec{V}_g \cdot \nabla (\zeta_g + f) + f_0 \frac{\partial \omega}{\partial p} \end{aligned} \quad (8)$$

From momentum equation, continuous equation, thermal equation, and EOS, we can derive following equation (I don't know what name it is)

$$\begin{aligned} \frac{\partial}{\partial t} \left(-\frac{\partial \Phi}{\partial p} \right) &= -\vec{V}_h \cdot \nabla \left(-\frac{\partial \Phi}{\partial p} \right) - \frac{R S_p \omega}{P} - \frac{R \dot{q}}{C_p P} \\ &= -\vec{V}_h \cdot \nabla \left(-\frac{\partial \Phi}{\partial p} \right) - \sigma \omega - Q \end{aligned} \quad (9)$$

Multiple (9) with $\frac{f_0}{\sigma} \frac{\partial}{\partial p}$ then plus (7) and multiple f_0 , we have

$$\left(\nabla_h^2 + \frac{f_0^2}{\sigma} \frac{\partial^2}{\partial p^2} \right) \chi = -f_0 \vec{V}_g \cdot \nabla (\zeta_g + f) + \frac{f_0^2}{\sigma} \frac{\partial}{\partial p} \left[\vec{V}_g \cdot \nabla \left(-\frac{\partial \Phi}{\partial p} \right) \right] - \frac{f_0^2}{\sigma} \frac{\partial Q}{\partial p}$$

During a non-adiabatic process, diabatic term is zero. However, during an adiabatic process, where Q represents heat, the preceding $\frac{f_0^2}{\sigma}$ is considered a constant to discuss the impact of heat on χ . If there is heat entering the system from the surroundings, i.e., $Q > 0$, due to the presence of the negative sign, χ will decrease, resulting in slower propagation speed of the westerly wave over time. Conversely, if there is heat leaving the system, $Q < 0$, χ will increase, leading to faster wave propagation over time.

2. With the omega equation shown below, please derive and interpret the physical meaning of the diabatic term.

$$\left(\nabla_h^2 + \frac{f_0^2}{\sigma} \frac{\partial^2}{\partial p^2} \right) \omega = \frac{f_0}{\sigma} \frac{\partial}{\partial p} \left[\vec{V}_g \cdot \nabla (\zeta_g + f) \right] + \frac{1}{\sigma} \nabla_h^2 \left[\vec{V}_g \cdot \nabla \left(-\frac{\partial \Phi}{\partial p} \right) \right] - \frac{1}{\sigma} \nabla_h^2 Q$$

Ans:

From the previous question, we modify (8) by $\zeta_g = \frac{\nabla_h^2 \Phi}{f_0}$, we obtain

$$\frac{\partial \zeta_g}{\partial t} = -\vec{V}_g \cdot \nabla \left(\frac{\nabla_h^2 \Phi}{f_0} + f \right) + f_0 \frac{\partial \omega}{\partial p} \quad (10)$$

Multiple (9) with $\frac{1}{f_0} \nabla_h^2$ and multiple (7) with $\partial \partial p$, then plus them together and divided by f_0 , we get

$$\left(\nabla_h^2 + \frac{f_0^2}{\sigma} \frac{\partial^2}{\partial p^2} \right) \omega = \frac{f_0}{\sigma} \frac{\partial}{\partial p} \left[\vec{V}_g \cdot \nabla (\zeta_g + f) \right] + \frac{1}{\sigma} \nabla_h^2 \left[\vec{V}_g \cdot \nabla \left(-\frac{\partial \Phi}{\partial p} \right) \right] - \frac{1}{\sigma} \nabla_h^2 Q$$

Similar to the previous question. During a non-adiabatic process, this term is zero. However, during an adiabatic process, where $\nabla^2 Q$ represents the heat advection term, and other factors are considered constants. If there is positive heat advection, i.e., $\nabla^2 Q > 0$, it will cause omega (ω) to decrease, resulting in a slower ascent speed. Conversely, the opposite is also true.

3. Under Boussinesq approximation, the vertical momentum equation can be written as

$$\frac{Dw}{dt} = -\frac{1}{\rho_0} \left(\frac{\partial p^*}{\partial z} + \rho^* g \right) + Fr(z)$$

which can be further simplified as

$$\frac{Dw}{dt} = -\frac{1}{\rho_0} \frac{\partial p^*}{\partial z} + \frac{\theta^*}{\theta_0} g + Fr(z)$$

where

$$-\frac{\rho^*}{\rho_0} g \cong \frac{\theta^*}{\theta_0} g$$

Verify this simplification.

Ans:

We know that

$$\begin{cases} \rho = \rho_0(z) + \rho^*(x, y, z) \\ \theta = \theta_0(z) + \theta^*(x, y, z) \end{cases}$$

From Possion equation

$$\theta = T \left(\frac{P_s}{P} \right)^\kappa = \frac{P}{\rho R} \left(\frac{P_s}{P} \right)^\kappa$$

Thus

$$\theta \sim \frac{1}{\rho}$$

Therefore, from above relation, we obtain

$$\frac{\theta^*}{\theta_0} = \frac{\theta - \theta_0}{\theta_0} = \frac{\frac{1}{\rho} - \frac{1}{\rho_0}}{\frac{1}{\rho_0}} = \frac{\rho_0}{\rho} - 1 = \frac{-\rho^*}{\rho} \cong -\frac{\rho^*}{\rho_0}$$

Since the variation of density is little near surface, $\rho \simeq \rho_0$

4. Prove that

$$\lim_{\gamma z \rightarrow 0} \frac{\bar{u}}{\bar{v}} = \lim_{\gamma z \rightarrow 0} \frac{e^{-\gamma z} \sin(\gamma z)}{1 - e^{-\gamma z} \cos(\gamma z)} = 1$$

where

$$\gamma = \sqrt{\frac{f}{2K_m}}$$

Ans:

$$\lim_{\gamma z \rightarrow 0} \left(\frac{e^{-\gamma z} \sin(\gamma z)}{1 - e^{-\gamma z} \cos(\gamma z)} \right)$$

Multiple by $e^{-\gamma z}$

$$= \lim_{\gamma z \rightarrow 0} \left(\frac{\sin(\gamma z)}{e^{\gamma z} - \cos(\gamma z)} \right)$$

Apply L'Hopital's Rule

$$= \lim_{\gamma z \rightarrow 0} \left(\frac{\cos(\gamma z)}{e^{\gamma z} + \sin(\gamma z)} \right)$$

Plug in the value $\gamma z = 0$

$$= \frac{\cos(0)}{e^0 + \sin(0)} = 1$$